

SeungYeon Back

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EDUCATION

Carnegie Mellon University, School of Computer Science, Pittsburgh, PA <i>Visiting Student, Software and Societal Systems Department (S3D)</i>	<i>Aug 2026 – present</i>
Hanyang University, Seoul, South Korea — M.S. in Artificial Intelligence <i>Advisor: Prof. Dong-Kyu Chae GPA: 4.25/4.5</i>	<i>Sep 2025 – present</i>
Hanyang University, Seoul, South Korea — B.S. in Mathematics <i>Minors: Biomedical Engineering, Big Data Convergence GPA: 3.97/4.5 (Cum Laude)</i>	<i>Mar 2021 – Aug 2025</i>

RESEARCH INTEREST

Large Language Model (LLM) Reasoning, LLM Agents, Retrieval-Augmented Generation (RAG)

RESEARCH EXPERIENCE

AdapDict: A Level- and Domain-Adaptive Educational Dictionary and Encyclopedia System <i>Manhwi Han*, SeungYeon Back*, Dong-Kyu Chae — CSCW Companion 2026 (Poster), Salt Lake City, UT</i> – Led the project end-to-end (idea, system design, web implementation) building a prompt-free LLM system generating level- and domain-adaptive explanations for learners. – Designed a Generator–Verifier loop with structured JSON outputs, cutting hallucinations by 50%; validated via a user study (n=41) with a 97.6% win rate and 85% learning-friendly rating.	<i>Jan 2026 – present</i>
GraphRAG-based Recommendation System <i>(in collaboration with LG Electronics)</i> <i>Graduate Researcher — Hanyang University, Seoul, KR</i> – Built a real-time ingestion pipeline for news and OTT content and designed a dual-graph framework aligning news and content graphs via shared entities. – Developed a multi-hop graph retrieval pipeline integrating LLM path-based reasoning for explainable candidate-content recommendations.	
An Empirical Study of the Clustering of Mountain Hiking GPS Trajectory Data <i>SeungYeon Back*, Manhwi Han*, Dong-Kyu Chae, Seoncheol Park — IEEE Access (Under Revision)</i> – Proposed a trajectory similarity measure with an ϵ-ball metric and adaptive spatiotemporal weights; applied hierarchical clustering to Worak Mountain GPS data to reveal latent hiking patterns.	

INTERNSHIP

Applied Statistics Lab, Hanyang University — Undergraduate Researcher – Built LSTM/GRU/BERT stock-prediction models in PyTorch and led peer study sessions on time-series modeling.	<i>Nov 2023 – Dec 2024</i>
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PROJECTS

Survival Analysis of Lung Cancer Patients — Cox PH & Random Survival Forest models	<i>09/2024 – 12/2024</i>
Stock Prediction with Sentiment Analysis — KoFinBERT + LSTM/GRU, +17% performance	<i>09/2024 – 12/2024</i>
Clustering Analysis of Corporate ESG Ratings — scraped & clustered firms by ESG	<i>07/2023 – 08/2023</i>

AWARDS & HONORS

Best Use of Vultr API (Sponsored Prize), HackCMU 2026	<i>2026</i>
Encouragement Award, Academic Conference	<i>2023 & 2024</i>

LEADERSHIP & ACTIVITIES

Mentoring Program, Volunteer Tutor — mentored high schoolers in math and time management	<i>07/2023 – 08/2023</i>
International Summer School, Universitat de Barcelona — Gastronomy & Cooking course	<i>07/2023 – 08/2023</i>
TEDxHYU, Scouting Director — recruited speakers and managed speaker relations	<i>09/2022 – 12/2022</i>
Student Council PR Team, Senior Manager — visual content & Instagram campaigns	<i>03/2021 – 12/2022</i>

SKILLS

Programming: Python, R **ML/LLM Frameworks:** PyTorch, HuggingFace Transformers